



Principal Component Analysis

Unsupervised learning: maximum variance, minimal reconstruction error, and probabilistic PCA

SPEAKER

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Reading: Bishop 12.1, 12.2

Outline

- 1 PCA via maximum variance
- 2 PCA via minimal reconstruction error
- 3 Probabilistic PCA

SECTION 1

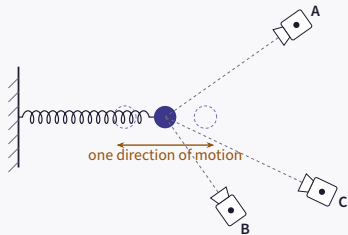
PCA via maximum variance

Unsupervised learning. Bishop 12.1.

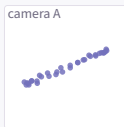


Motivation: a ball on a spring

- A ball on a spring moves back and forth along one direction, which we do not know
- Three cameras at arbitrary angles record its position: $\mathbf{x}_t = (x_A, y_A, x_B, y_B, x_C, y_C) \in \mathbb{R}^6$



what the cameras record: two coordinates each, at every time step



(x_A, y_A)



(x_B, y_B)



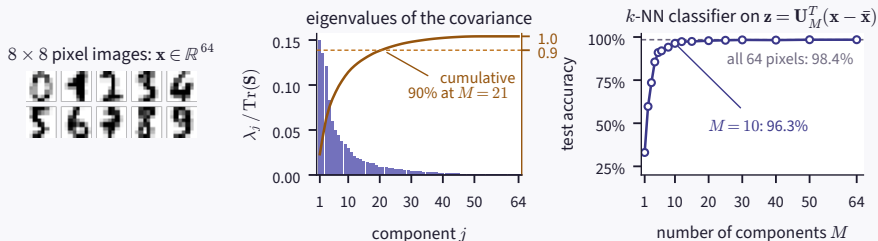
(x_C, y_C)

after J. Shlens, *A tutorial on principal component analysis* (2014)

- 6 numbers per time step, but 1 degree of freedom. Which direction in $\mathbb{R}^6$ is the spring?
The one along which the data varies most: $\mathbf{u}_1 \in \mathbb{R}^6$, with $z = \mathbf{u}_1^T \mathbf{x}$ the displacement

Motivation: high-dimensional data

- In ML the inputs are often high-dimensional: an 8×8 image is already a vector $\mathbf{x} \in \mathbb{R}^{64}$
- The pixels are strongly correlated: most of the variance lies in a few directions



- $M = 10$ features classify almost as well as all 64 pixels: **fewer parameters**, less data needed

Continuous latent space

- **Dimensionality reduction:** model the data in a low dim. space
- Example: take one grey-scale image of “3” and make multiple copies by translation and rotation



Figure: Synthetic “3” dataset (Bishop 12.1)

- Pixel space dimension: 100×100 pixels
- Latent space dimension: $3 = 2$ (translations) + 1 rotation
- From the 3 latent variables we could generate all 100×100 pixels!

Example continued

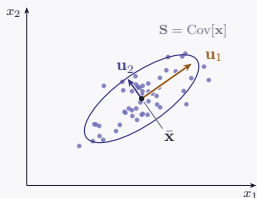
- A more realistic dataset of images will have more degrees of freedom in the latent space, such as:
 - Scaling
 - Digits from 0–9
 - Colors
 - Different hand-writing styles
 - Etc.

... but still much fewer than 100×100 !
- In this example, the latent subspace represents a non-linear transformation of the images
- We first study linear latent spaces with PCA and later consider generalizations to the non-linear case

Principal Component Analysis (PCA)

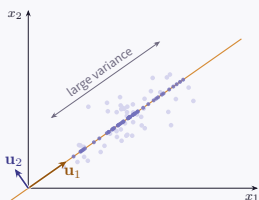
- Find a linear projection of the data such that the variance in the projected space is maximal
- PCA captures the axes of maximal variation in the data, called **principal components**

1 · THE IDEA



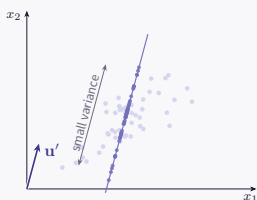
Analyze $S = \text{Cov}[\mathbf{x}]$, the covariance around $\bar{\mathbf{x}}$.

2 · PROJECT ONTO $\mathbf{u}_1$



The projected data has large variance.

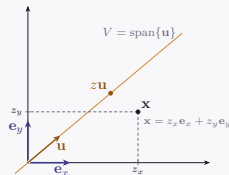
3 · ANOTHER DIRECTION $\mathbf{u}'$



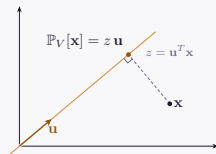
Projected onto any other direction, the variance is smaller: PCA picks $\mathbf{u}_1$.

Recall orthonormal projections

- ① Let $\mathbf{u} \in \mathbb{R}^D$ be a **normalized vector** ($\|\mathbf{u}\| = 1$). A single vector spans a 1-dimensional vector space $V = \text{span}\{\mathbf{u}\} = \{z\mathbf{u} \mid z \in \mathbb{R}\}$
- ② An **orthonormal basis** consists of
 - a collection of normalized vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_d\}$
 - all orthogonal: $\mathbf{u}_i^T \mathbf{u}_j = 1$ if $i = j$, 0 otherwise
 and spans a d -dimensional vector space $V = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_d\} = \{\sum_i z_i \mathbf{u}_i \mid z_i \in \mathbb{R}\}$
- ③ A vector $\mathbf{x}$ can be projected to such a space via an **orthonormal projection** $\mathbb{P}_V[\mathbf{x}] = \sum_{i=1}^d (\mathbf{u}_i^T \mathbf{x}) \mathbf{u}_i$
 $z_i = \mathbf{u}_i^T \mathbf{x}$: “coordinates relative to $\mathbf{u}_i$ ”



① $V = \text{span}\{\mathbf{u}\}$ ② $\mathbf{x} = z_x \mathbf{e}_x + z_y \mathbf{e}_y$



③ $\mathbb{P}_V[\mathbf{x}] = z\mathbf{u}$ with $z = \mathbf{u}^T \mathbf{x}$

Principal Component Analysis (PCA)

- Data: $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, $\mathbf{x}_n \in \mathbb{R}^D$
- **Goal: project data into a $M < D$ dimensional space while maximizing the variance of the projected data**
- M is given
- Mean and covariance defined by

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n, \quad \mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- $\mathbf{S}$ is symmetric and positive *semi*-definite

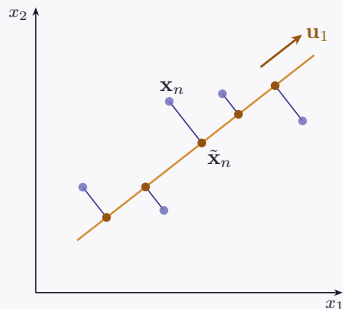


Figure: Maximizing variance of projections
(Bishop 12.2)

1D Projection

- 1 Project data into the first latent dimension by a vector $\mathbf{u}_1 \in \mathbb{R}^D$
- 2 The projection gives the scalar $z_1 = \mathbf{u}_1^T \mathbf{x}_n$ (a single number); its mean is $\bar{z}_1 = \mathbf{u}_1^T \bar{\mathbf{x}}$
- 3 We only need its direction, so normalize this component: $\|\mathbf{u}_1\|^2 = \mathbf{u}_1^T \mathbf{u}_1 = 1$ constraint!

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$$\text{Var}[z_1] = \mathbb{E}[(z_1 - \bar{z}_1)^2] = \frac{1}{N} \sum_{n=1}^N (\mathbf{u}_1^T \mathbf{x}_n - \mathbf{u}_1^T \bar{\mathbf{x}})^2 = \frac{1}{N} \sum_{n=1}^N (\mathbf{u}_1^T (\mathbf{x}_n - \bar{\mathbf{x}}))^2$$

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Maximizing the variance of 1 component

- Solve $\arg \max_{\mathbf{u}_1} \underbrace{\mathbf{u}_1^T \mathbf{S} \mathbf{u}_1}_{f(\mathbf{u}_1)}$ subject to $\underbrace{\mathbf{u}_1^T \mathbf{u}_1 = 1}_{g(\mathbf{u}_1) = 1}$

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- Method of Lagrange multipliers
 - Define Lagrangian

$$L(\mathbf{u}_1, \lambda_1) = \mathbf{u}_1^T \mathbf{S} \mathbf{u}_1 + \lambda_1 (1 - \mathbf{u}_1^T \mathbf{u}_1)$$

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- Solving for $\mathbf{u}_1$ means

- **We need to solve the eigensystem**

- So $\mathbf{u}_1$ and λ_1 are respectively an eigenvector and eigenvalue of $\mathbf{S} \in \mathbb{R}^{D \times D}$!

$$\begin{aligned} L(\mathbf{u}_1, \lambda_1) &= \mathbf{u}_1^T \mathbf{S} \mathbf{u}_1 + \lambda_1 (1 - \mathbf{u}_1^T \mathbf{u}_1) \\ \frac{\partial}{\partial \mathbf{u}_1} L(\mathbf{u}_1, \lambda_1) &= 2\mathbf{S} \mathbf{u}_1 - 2\lambda_1 \mathbf{u}_1 = 0 \\ &\Rightarrow \mathbf{S} \mathbf{u}_1 = \lambda_1 \mathbf{u}_1 \end{aligned}$$

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- The $\mathbf{u}_1$ is called a **principal component**.

- The variance of the projected data is $\mathbf{u}_1^T \mathbf{S} \mathbf{u}_1 = \lambda_1 \mathbf{u}_1^T \mathbf{u}_1 = \lambda_1$

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- **Maximizing variance means we search for the eigenvector with the largest eigenvalue**

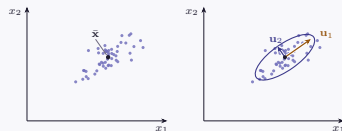
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$$\Rightarrow \mathbf{S} \mathbf{u}_1 = \lambda_1 \mathbf{u}_1$$

PCA via maximum variance

- We repeat the procedure for M orthogonal vectors and get a projection defined by $\mathbf{U}_M = [\mathbf{u}_1, \dots, \mathbf{u}_M] \in \mathbb{R}^{D \times M}$
- PCA:
 - Compute $\bar{\mathbf{x}}$
 - Compute covariance $\mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$
 - Compute the M eigenvectors of $\mathbf{S}$ with largest eigenvalue: $\mathbf{U}_M$
 - The **projection** to the latent vector then is $\mathbf{z} = \mathbf{U}_M^T (\mathbf{x} - \bar{\mathbf{x}})$
- $\mathbf{U}_M$ contains the M eigenvectors of $\mathbf{S}$, the **principal components**. The eigenvalues are $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_M$
- The matrix $\mathbf{S}$ is positive semi-definite, thus $\forall j : \lambda_j \geq 0$
- The (total) variance of the projected data is $\text{Tr}[\text{Cov}[\mathbf{z}]] = \sum_{j=1}^M \lambda_j$

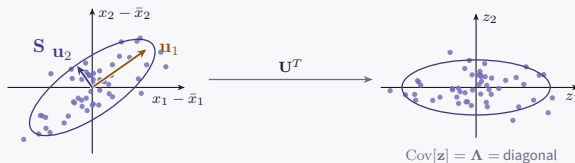


Reminder: eigen-decomposition

- When the matrix is **symmetric positive semi-definite**:

$$\mathbf{S} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T \quad \text{with} \quad \mathbf{\Lambda} = \text{diag}\{\lambda_1, \dots, \lambda_D\} \quad \text{change of basis!}$$

- The eigenvectors are **orthonormal** and are stored in $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_D]$
- All eigenvalues are **non-negative** and are the elements of the **diagonal matrix $\mathbf{\Lambda}$**
- Total variance given by $\text{Tr}(\mathbf{S}) = \text{Tr}(\mathbf{U} \mathbf{\Lambda} \mathbf{U}^T) = \text{Tr}(\mathbf{\Lambda}) = \sum_{i=1}^D \lambda_i$



In projected space the points $\mathbf{z}$ are **decorrelated**: independent random variables, $\text{Cov}[\mathbf{z}] = \mathbf{\Lambda} = \text{diagonal}$.

Getting the eigenvectors in practice

- Full eigenvalue decomposition is expensive: $O(D^3)$
- Only need up to the M^{th} component: $O(MD^2)$
- In python:

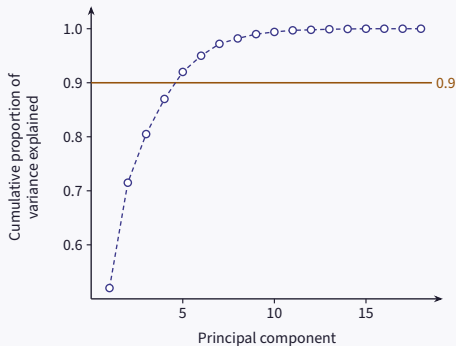
```
M = 10  
S = np.cov(X)  
Um, Lm, Vm = scipy.sparse.linalg.svds(S, k=M)
```

$\mathbf{S} = \mathbf{U}_M \mathbf{L}_M \mathbf{V}_M$: singular value decomposition

For symmetric positive definite matrices such as $\mathbf{S}$, the SVD decomposition is equivalent to the eigen-decomposition

How to choose M ?

- We can measure the discarded variance
- For example to preserve 90% of the variance, pick M such that



**THE PROPORTION OF EXPLAINED
VARIANCE**

$$\frac{\sum_{j=1}^M \lambda_j}{\text{Tr}(\mathbf{S})} = \frac{\sum_{j=1}^M \lambda_j}{\sum_{i=1}^D \lambda_i} > 0.9$$

Applications: dimensionality reduction

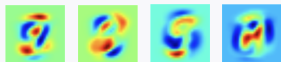
- When data is defined in high dimension (large D) we want to project down to lower dimension because:
 - Reduce time and storage space required
 - For classification/regression: our model **will have less parameters**, thus we need less data points for learning
- Other methods (not covered): **feature selection**. PCA is known as a **feature extraction** method instead.

Applications: 2D visualization (MNIST)



MNIST: 28×28 pixels

EIGENVECTORS

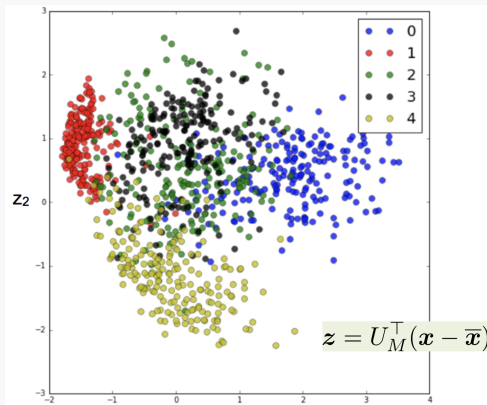


u_1

u_2

u_3

u_4



Feature decorrelation

- Good side effect of PCA: features have **no correlation** in the projected space.
- The covariance matrix of the projected data is **diagonal**

$$\frac{1}{N} \sum_{n=1}^N \mathbf{z}_n \mathbf{z}_n^T = \frac{1}{N} \sum_{n=1}^N \mathbf{U}_M^T (\mathbf{x}_n - \bar{\mathbf{x}}) (\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{U}_M$$

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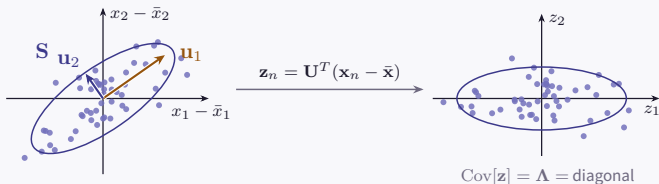
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Feature decorrelation

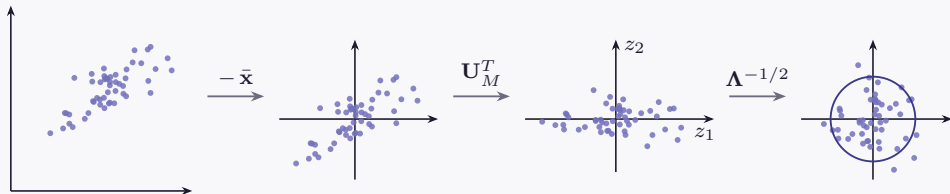
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Applications: whitening (or sphering)

- Before applying learning algorithms we usually do some pre-processing:
 - e.g. **standardization**: subtract the mean and divide by the standard deviation
- With PCA we can **whiten** the data, one step more:
 - **Centre** and **de-correlate** the features: $\mathbf{z} = \mathbf{U}_M^T (\mathbf{x} - \bar{\mathbf{x}})$
 - Cast features to **unit standard deviation** by rescaling: $\mathbf{z} = \mathbf{\Lambda}^{-1/2} \mathbf{U}_M^T (\mathbf{x} - \bar{\mathbf{x}})$



SECTION 2

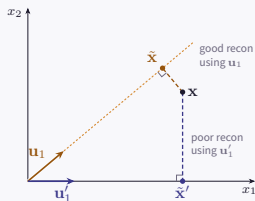
PCA via minimal reconstruction error

Unsupervised learning. Principal Component
Analysis: reconstruction error minimization. Bishop
12.1.2, 12.1.3.



PCA via minimal reconstruction error

- Same method, alternative view to the maximal variance viewpoint
- PCA can be obtained by minimizing the reconstruction error. Find a transformation that minimizes:



$$\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|$$

with the reconstruction $\tilde{\mathbf{x}}_n = \mathbb{P}_{\mathbf{U}_M}[\mathbf{x}_n - \bar{\mathbf{x}}] + \bar{\mathbf{x}} = \mathbf{U}_M \mathbf{z}_n + \bar{\mathbf{x}}$

- Represent the points in a different orthonormal basis (unknown for now):

$$\{\mathbf{u}_i : \mathbf{u}_i^T \mathbf{u}_i = 1\}_{i=1}^D, \quad \mathbf{u}_i^T \mathbf{u}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

PCA via minimal reconstruction error

- In the new basis $\mathbf{x}_n = \sum_{i=1}^D \alpha_{ni} \mathbf{u}_i$

PCA via minimal reconstruction error

- In the new basis $\mathbf{x}_n = \sum_{i=1}^D \alpha_{ni} \mathbf{u}_i$
- The coefficients are found by multiplying with $\mathbf{u}_j^T$ on both sides:

$$\mathbf{u}_j^T \mathbf{x}_n = \sum_{i=1}^D \alpha_{ni} \underbrace{\mathbf{u}_j^T \mathbf{u}_i}_{= 1 \text{ if } i = j, 0 \text{ otherwise}} = \alpha_{nj} \quad \implies \quad \alpha_{ni} = \mathbf{x}_n^T \mathbf{u}_i$$

PCA via minimal reconstruction error

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- The coefficients are found by multiplying with $\mathbf{u}_j^T$ on both sides:

$$\mathbf{u}_j^T \mathbf{x}_n = \sum_{i=1}^D \alpha_{ni} \underbrace{\mathbf{u}_j^T \mathbf{u}_i}_{= 1 \text{ if } i = j, 0 \text{ otherwise}} = \alpha_{nj} \implies \alpha_{ni} = \mathbf{x}_n^T \mathbf{u}_i$$

- Therefore... we get the **orthonormal projection** onto the new basis: $\mathbf{x}_n = \sum_{i=1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i = \mathbb{P}_{\mathbf{U}}[\mathbf{x}_n]$

PCA via minimal reconstruction error

- In the new basis $\mathbf{x}_n = \sum_{i=1}^D \alpha_{ni} \mathbf{u}_i$
- The coefficients are found by multiplying with $\mathbf{u}_j^T$ on both sides:

$$\mathbf{u}_j^T \mathbf{x}_n = \sum_{i=1}^D \alpha_{ni} \underbrace{\mathbf{u}_j^T \mathbf{u}_i}_{= 1 \text{ if } i = j, 0 \text{ otherwise}} = \alpha_{nj} \implies \alpha_{ni} = \mathbf{x}_n^T \mathbf{u}_i$$

- Therefore... we get the **orthonormal projection** onto the new basis: $\mathbf{x}_n = \sum_{i=1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i = \mathbb{P}_{\mathbf{U}}[\mathbf{x}_n]$
- For reconstruction, we use the first **M** elements of the basis and a shared offset for the remaining dimensions

$$\tilde{\mathbf{x}}_n = \sum_{i=1}^M (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i + \sum_{i=M+1}^D b_i \mathbf{u}_i$$

$\mathbf{x}_n^T \mathbf{u}_i = \alpha_{ni}$: data dependent coefficient

b_i : shared for all data points

PCA via minimal reconstruction error

- The difference/error is then given by

$$\mathbf{x}_n - \tilde{\mathbf{x}}_n = \overbrace{\sum_{i=1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i}^{\mathbf{x}_n} - \overbrace{\sum_{i=1}^M (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - \sum_{i=M+1}^D b_i \mathbf{u}_i}^{\tilde{\mathbf{x}}_n}$$

PCA via minimal reconstruction error

- The difference/error is then given by

$$\begin{aligned}\mathbf{x}_n - \tilde{\mathbf{x}}_n &= \overbrace{\sum_{i=1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i}^{\mathbf{x}_n} - \overbrace{\sum_{i=1}^M (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - \sum_{i=M+1}^D b_i \mathbf{u}_i}^{\tilde{\mathbf{x}}_n} \\ &= \sum_{i=M+1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - \sum_{i=M+1}^D b_i \mathbf{u}_i\end{aligned}$$

PCA via minimal reconstruction error

- The difference/error is then given by

$$\begin{aligned}
 \mathbf{x}_n - \tilde{\mathbf{x}}_n &= \overbrace{\sum_{i=1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i}^{\mathbf{x}_n} - \overbrace{\sum_{i=1}^M (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - \sum_{i=M+1}^D b_i \mathbf{u}_i}^{\tilde{\mathbf{x}}_n} \\
 &= \sum_{i=M+1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - \sum_{i=M+1}^D b_i \mathbf{u}_i \\
 &= \sum_{i=M+1}^D (\mathbf{x}_n^T \mathbf{u}_i - b_i) \mathbf{u}_i
 \end{aligned}$$

PCA via minimal reconstruction error

- We have to **minimize for both $\mathbf{u}_i$ and b_i**

$$\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - b_i \mathbf{u}_i \right\|^2$$

PCA via minimal reconstruction error

- We have to **minimize for both $\mathbf{u}_i$ and b_i**

$$\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - b_i \mathbf{u}_i \right\|^2$$

- Solution for b_i is given by $b_i = \bar{\mathbf{x}}^T \mathbf{u}_i$ $\frac{\partial}{\partial b_i}(\dots) = 0 \rightarrow b_i$

PCA via minimal reconstruction error

- We have to **minimize for both $\mathbf{u}_i$ and b_i**

$$\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D (\mathbf{x}_n^T \mathbf{u}_i) \mathbf{u}_i - b_i \mathbf{u}_i \right\|^2$$

- Solution for b_i is given by $b_i = \bar{\mathbf{x}}^T \mathbf{u}_i$ $\frac{\partial}{\partial b_i}(\dots) = 0 \rightarrow b_i$
- We are left to solve for $\mathbf{u}_i$, minimizing

$$\frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i) \mathbf{u}_i \right\|^2$$

PCA via minimal reconstruction error

$$\frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i) \mathbf{u}_i \right\|^2$$

PCA via minimal reconstruction error

$$\begin{aligned} & \frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i) \mathbf{u}_i \right\|^2 \\ &= \frac{1}{N} \sum_{n=1}^N \left(\sum_{i=M+1}^D \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i)}_{\text{scalar}} \underbrace{\mathbf{u}_i}_{\text{vector}} \right)^T \left(\sum_{j=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_j) \mathbf{u}_j \right) \end{aligned}$$

PCA via minimal reconstruction error

$$\begin{aligned}
 & \frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i) \mathbf{u}_i \right\|^2 \\
 &= \frac{1}{N} \sum_{n=1}^N \left(\sum_{i=M+1}^D \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i)}_{\text{scalar}} \underbrace{\mathbf{u}_i}_{\text{vector}} \right)^T \left(\sum_{j=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_j) \mathbf{u}_j \right) \\
 &= \frac{1}{N} \sum_{n=1}^N \sum_{i=M+1}^D \sum_{j=M+1}^D \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i)}_{\mathbf{a}^T \mathbf{b}} \underbrace{\mathbf{u}_i^T \mathbf{u}_j}_{= 1 \text{ if } i = j, 0 \text{ otherwise}} \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_j)}_{\mathbf{b}^T \mathbf{a}}
 \end{aligned}$$

PCA via minimal reconstruction error

$$\begin{aligned}
 & \frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i) \mathbf{u}_i \right\|^2 \\
 &= \frac{1}{N} \sum_{n=1}^N \left(\sum_{i=M+1}^D \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i)}_{\text{scalar}} \underbrace{\mathbf{u}_i}_{\text{vector}} \right)^T \left(\sum_{j=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_j) \mathbf{u}_j \right) \\
 &= \frac{1}{N} \sum_{n=1}^N \sum_{i=M+1}^D \sum_{j=M+1}^D \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i)}_{\mathbf{a}^T \mathbf{b}} \underbrace{\mathbf{u}_i^T \mathbf{u}_j}_{= 1 \text{ if } i=j, 0 \text{ otherwise}} \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_j)}_{\mathbf{b}^T \mathbf{a}} \\
 &= \frac{1}{N} \sum_{n=1}^N \sum_{i=M+1}^D \mathbf{u}_i^T (\mathbf{x}_n - \bar{\mathbf{x}}) (\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i = \sum_{i=M+1}^D \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i
 \end{aligned}$$

PCA via minimal reconstruction error

$$\begin{aligned}
 & \frac{1}{N} \sum_{n=1}^N \left\| \sum_{i=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i) \mathbf{u}_i \right\|^2 \\
 &= \frac{1}{N} \sum_{n=1}^N \left(\sum_{i=M+1}^D \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i)}_{\text{scalar}} \underbrace{\mathbf{u}_i}_{\text{vector}} \right)^T \left(\sum_{j=M+1}^D ((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_j) \mathbf{u}_j \right) \\
 &= \frac{1}{N} \sum_{n=1}^N \sum_{i=M+1}^D \sum_{j=M+1}^D \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i)}_{\mathbf{a}^T \mathbf{b}} \underbrace{\mathbf{u}_i^T \mathbf{u}_j}_{= 1 \text{ if } i = j, 0 \text{ otherwise}} \underbrace{((\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_j)}_{\mathbf{b}^T \mathbf{a}} \\
 &= \frac{1}{N} \sum_{n=1}^N \sum_{i=M+1}^D \mathbf{u}_i^T (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T \mathbf{u}_i = \sum_{i=M+1}^D \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i
 \end{aligned}$$

PCA via minimal reconstruction error

- We got: $\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \sum_{i=M+1}^D \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i = \sum_{i=M+1}^D \lambda_i$ discarded variance

PCA via minimal reconstruction error

- We got: $\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \sum_{i=M+1}^D \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i = \sum_{i=M+1}^D \lambda_i$ discarded variance
- Solve for $\mathbf{u}_i$ under constraint $\mathbf{u}_i^T \mathbf{u}_i = 1$ (else we get $\mathbf{u}_i = \mathbf{0}$)

PCA via minimal reconstruction error

- We got: $\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \sum_{i=M+1}^D \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i = \sum_{i=M+1}^D \lambda_i$ discarded variance
- Solve for $\mathbf{u}_i$ under constraint $\mathbf{u}_i^T \mathbf{u}_i = 1$ (else we get $\mathbf{u}_i = \mathbf{0}$)
- Method of Lagrange multipliers $\rightarrow$ Solve eigensystem

PCA via minimal reconstruction error

- We got: $\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \sum_{i=M+1}^D \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i = \sum_{i=M+1}^D \lambda_i$ discarded variance
- Solve for $\mathbf{u}_i$ under constraint $\mathbf{u}_i^T \mathbf{u}_i = 1$ (else we get $\mathbf{u}_i = \mathbf{0}$)
- Method of Lagrange multipliers $\rightarrow$ Solve eigensystem
- **Find the largest M eigenvectors and values, such that the remaining $(D - M)$ are the smallest**

PCA via minimal reconstruction error

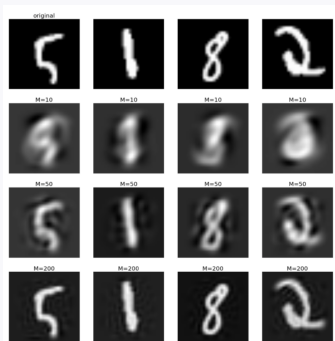
- We got: $\frac{1}{N} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|^2 = \sum_{i=M+1}^D \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i = \sum_{i=M+1}^D \lambda_i$ discarded variance
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- Method of Lagrange multipliers $\rightarrow$ Solve eigensystem
- **Find the largest M eigenvectors and values, such that the remaining $(D - M)$ are the smallest**

min reconstruction error = max projected variance

“using the set of principal components with least discarded variance!”

Applications: compression (MNIST)

EIGENVECTORS

 u_1  u_2  u_3  u_4 RECONSTRUCTIONS $\tilde{x} = U_M z + \bar{x}$ 

original

 $M = 10$ $M = 50$ $M = 200$

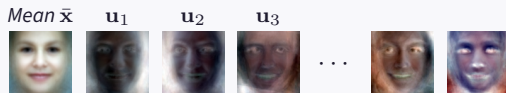
Applications: compression (eigenfaces/UTKFace)

Dataset:



Task: Compress image or feature selection for face recognition

Method: Expand along principal components (PCA)



Result:

Original



$$\approx \bar{\mathbf{x}} + \sum_{i=1}^M z_{ni} \mathbf{u}_i$$

$M = 1$



$M = 10$



$M = 50$



$M = 150$



SECTION 3

Probabilistic PCA

Unsupervised learning. PCA as the maximum likelihood solution of a probabilistic latent variable model, which makes it generative. Bishop 12.2.1.



Probabilistic PCA

- Probabilistic view of PCA:
- Learn it via maximum likelihood
- (Third) alternative view of PCA
- Both latent and observed variables are Gaussian

Continuous latent variable model

- Data: $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, $\mathbf{x}_n \in \mathbb{R}^D$
- Goal: learn a $M < D$ **continuous latent space** by maximizing the likelihood of the probabilistic model, M given
- Recall the continuous latent variable model a “continuous Gaussian mixture model”

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p(\mathbf{x} | \mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$

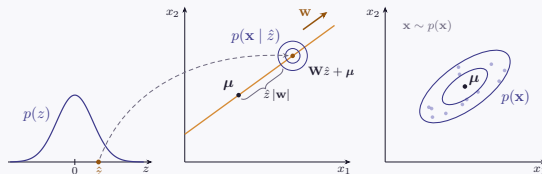


Figure: Bishop 12.9

PPCA modeling assumptions

- The **generative model** works as follows

$$\mathbf{x} = \mathbf{W} \mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}$$

- With $\boldsymbol{\mu} \in \mathbb{R}^D$ and the **continuous latent variable** $\mathbf{z} \in \mathbb{R}^M$ with Gaussian **prior**

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})$$

- Matrix $\mathbf{W} \in \mathbb{R}^{D \times M}$ transforms the latent variables into observed variables.
- The independent noise is also Gaussian

$$p(\boldsymbol{\epsilon}) = \mathcal{N}(\boldsymbol{\epsilon} \mid \mathbf{0}, \sigma^2 \mathbf{I})$$

$\mathbf{W}$, $\boldsymbol{\mu}$ and σ are the model parameters that we want to recover.

It follows...

- The conditional distribution of the observed variable

$$p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$$

It follows...

- The conditional distribution of the observed variable

$$p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

- **The marginal $p(\mathbf{x})$ is also a Gaussian (Bishop Ch. 2.3)**

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) \, d\mathbf{z} = \int p(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z}) \, d\mathbf{z} = \mathcal{N}(\mathbf{x} \mid \underbrace{?}_{\mathbb{E}[\mathbf{x}]}, \underbrace{?}_{\text{Cov}[\mathbf{x}]})$$

It follows...

- The conditional distribution of the observed variable

$$p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

- **The marginal $p(\mathbf{x})$ is also a Gaussian (Bishop Ch. 2.3)**

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) \, d\mathbf{z} = \int p(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z}) \, d\mathbf{z} = \mathcal{N}(\mathbf{x} \mid \underbrace{?}_{\mathbb{E}[\mathbf{x}]}, \underbrace{?}_{\text{Cov}[\mathbf{x}]})$$

- The expected value for $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\mathbb{E}[\mathbf{x}] = \mathbb{E}[\mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}]$$

It follows...

- The conditional distribution of the observed variable

$$p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

- The marginal $p(\mathbf{x})$ is also a Gaussian (Bishop Ch. 2.3)**

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) \, d\mathbf{z} = \int p(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z}) \, d\mathbf{z} = \mathcal{N}(\mathbf{x} \mid \underbrace{?}_{\mathbb{E}[\mathbf{x}]}, \underbrace{?}_{\text{Cov}[\mathbf{x}]})$$

- The expected value for $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned} \mathbb{E}[\mathbf{x}] &= \mathbb{E}[\mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}] \\ &= \mathbf{W} \underbrace{\mathbb{E}[\mathbf{z}]}_{=0} + \boldsymbol{\mu} + \underbrace{\mathbb{E}[\boldsymbol{\epsilon}]}_{=0} = \boldsymbol{\mu} \end{aligned}$$

It follows...

- The conditional distribution of the observed variable

$$p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I})$$

- The marginal $p(\mathbf{x})$ is also a Gaussian (Bishop Ch. 2.3)**

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) \, d\mathbf{z} = \int p(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z}) \, d\mathbf{z} = \mathcal{N}(\mathbf{x} \mid \underbrace{?}_{\mathbb{E}[\mathbf{x}]}, \underbrace{?}_{\text{Cov}[\mathbf{x}]})$$

- The expected value for $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned} \mathbb{E}[\mathbf{x}] &= \mathbb{E}[\mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}] \\ &= \mathbf{W} \underbrace{\mathbb{E}[\mathbf{z}]}_{=0} + \boldsymbol{\mu} + \underbrace{\mathbb{E}[\boldsymbol{\epsilon}]}_{=0} = \boldsymbol{\mu} \end{aligned}$$

The mean of the marginal Gaussian will thus be given by model parameter $\boldsymbol{\mu}$.

It follows...

- The covariance of $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\text{Cov}[\mathbf{x}] = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T]$$

It follows...

- The covariance of $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned}\text{Cov}[\mathbf{x}] &= \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \\ &= \mathbb{E}[(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})^T]\end{aligned}$$

It follows...

- The covariance of $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned}\text{Cov}[\mathbf{x}] &= \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \\ &= \mathbb{E}[(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})^T] \\ &= \mathbb{E}[\mathbf{W}\mathbf{z}\mathbf{z}^T\mathbf{W}^T + 2\mathbf{W}\mathbf{z}\boldsymbol{\epsilon}^T + \boldsymbol{\epsilon}\boldsymbol{\epsilon}^T]\end{aligned}$$

It follows...

- The covariance of $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned}
 \text{Cov}[\mathbf{x}] &= \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \\
 &= \mathbb{E}[(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})^T] \\
 &= \mathbb{E}[\mathbf{W}\mathbf{z}\mathbf{z}^T\mathbf{W}^T + 2\mathbf{W}\mathbf{z}\boldsymbol{\epsilon}^T + \boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] \\
 &= \mathbf{W} \mathbb{E}[\mathbf{z}\mathbf{z}^T] \mathbf{W}^T + 2\mathbf{W} \underbrace{\mathbb{E}[\mathbf{z}\boldsymbol{\epsilon}^T]}_{=0} + \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T]
 \end{aligned}$$

It follows...

- The covariance of $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned}
 \text{Cov}[\mathbf{x}] &= \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \\
 &= \mathbb{E}[(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})^T] \\
 &= \mathbb{E}[\mathbf{W}\mathbf{z}\mathbf{z}^T\mathbf{W}^T + 2\mathbf{W}\mathbf{z}\boldsymbol{\epsilon}^T + \boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] \\
 &= \mathbf{W} \mathbb{E}[\mathbf{z}\mathbf{z}^T] \mathbf{W}^T + 2\mathbf{W} \underbrace{\mathbb{E}[\mathbf{z}\boldsymbol{\epsilon}^T]}_{=0} + \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T]
 \end{aligned}$$

$$\mathbf{x} - \boldsymbol{\mu} = \mathbf{W}\mathbf{z} + \boldsymbol{\epsilon}$$

$$\text{Cov}[\mathbf{z}] = \mathbb{E}[\mathbf{z}\mathbf{z}^T] - \cancel{\mathbb{E}[\mathbf{z}]\mathbb{E}[\mathbf{z}]^T} = \mathbf{I}$$

$$\text{Cov}[\boldsymbol{\epsilon}] = \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] - \cancel{\mathbb{E}[\boldsymbol{\epsilon}]\mathbb{E}[\boldsymbol{\epsilon}]^T} = \sigma^2\mathbf{I}$$

$$\text{Cov}[\mathbf{z}, \boldsymbol{\epsilon}] = \mathbb{E}[\mathbf{z}\boldsymbol{\epsilon}^T] - \cancel{\mathbb{E}[\mathbf{z}]\mathbb{E}[\boldsymbol{\epsilon}]^T} = \mathbf{0}$$

It follows...

- The covariance of $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned}
 \text{Cov}[\mathbf{x}] &= \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \\
 &= \mathbb{E}[(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})^T] \\
 &= \mathbb{E}[\mathbf{W}\mathbf{z}\mathbf{z}^T\mathbf{W}^T + 2\mathbf{W}\mathbf{z}\boldsymbol{\epsilon}^T + \boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] \\
 &= \mathbf{W} \mathbb{E}[\mathbf{z}\mathbf{z}^T] \mathbf{W}^T + 2\mathbf{W} \underbrace{\mathbb{E}[\mathbf{z}\boldsymbol{\epsilon}^T]}_{=0} + \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] \\
 &= \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I} = \mathbf{C}
 \end{aligned}$$

$$\mathbf{x} - \boldsymbol{\mu} = \mathbf{W}\mathbf{z} + \boldsymbol{\epsilon}$$

$$\text{Cov}[\mathbf{z}] = \mathbb{E}[\mathbf{z}\mathbf{z}^T] - \cancel{\mathbb{E}[\mathbf{z}]\mathbb{E}[\mathbf{z}]^T} = \mathbf{I}$$

$$\text{Cov}[\boldsymbol{\epsilon}] = \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] - \cancel{\mathbb{E}[\boldsymbol{\epsilon}]\mathbb{E}[\boldsymbol{\epsilon}]^T} = \sigma^2\mathbf{I}$$

$$\text{Cov}[\mathbf{z}, \boldsymbol{\epsilon}] = \mathbb{E}[\mathbf{z}\boldsymbol{\epsilon}^T] - \cancel{\mathbb{E}[\mathbf{z}]\mathbb{E}[\boldsymbol{\epsilon}]^T} = \mathbf{0}$$

It follows...

- The covariance of $\mathbf{x} \sim p(\mathbf{x})$ is given by

$$\begin{aligned}
 \text{Cov}[\mathbf{x}] &= \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^T] \\
 &= \mathbb{E}[(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})(\mathbf{W}\mathbf{z} + \boldsymbol{\epsilon})^T] \\
 &= \mathbb{E}[\mathbf{W}\mathbf{z}\mathbf{z}^T\mathbf{W}^T + 2\mathbf{W}\mathbf{z}\boldsymbol{\epsilon}^T + \boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] \\
 &= \mathbf{W} \mathbb{E}[\mathbf{z}\mathbf{z}^T] \mathbf{W}^T + 2\mathbf{W} \underbrace{\mathbb{E}[\mathbf{z}\boldsymbol{\epsilon}^T]}_{=0} + \mathbb{E}[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T] \\
 &= \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I} = \mathbf{C}
 \end{aligned}$$

- Therefore $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{C})$

And its covariance $\mathbf{C}$ will depend on model parameters $\mathbf{W}, \sigma$ via $\mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}$.

$$\mathbf{x} - \boldsymbol{\mu} = \mathbf{W}\mathbf{z} + \boldsymbol{\epsilon}$$

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Probabilistic PCA in a picture

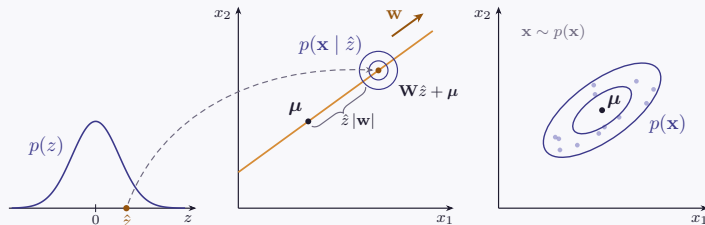


Figure: Bishop 12.9

- Prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})$
- Likelihood $p(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x} \mid \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I})$
- Marginal $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \mathbf{C})$ with $\mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2 \mathbf{I}$

The log-likelihood

$\mathbf{x} = \mathbf{W}\mathbf{z} + \boldsymbol{\mu} + \boldsymbol{\epsilon}$, $\mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}$. Now the likelihood is defined, let's use it to find the model parameters via ML.

$$\ln p(\mathbf{X} \mid \boldsymbol{\mu}, \mathbf{W}, \sigma^2) = \sum_{n=1}^N \ln p(\mathbf{x}_n \mid \boldsymbol{\mu}, \mathbf{W}, \sigma^2) = \sum_{n=1}^N \ln \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}, \mathbf{C})$$

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- Once again, take its derivative w.r.t. the parameter of interest, set it to zero, and solve it. For the mean:

$$\sum_{n=1}^N \mathbf{C}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}) = 0 \quad \implies \quad \boldsymbol{\mu} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$$

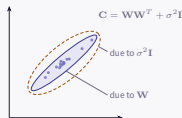
PPCA has closed-form solutions

- Let $\mathbf{S}$ be the sample covariance, as defined in PCA, with its eigenvectors and eigenvalues given
- The optimal parameters for the maximal log-likelihood are

$$\boldsymbol{\mu} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$$

$$\sigma^2 = \frac{1}{D - M} \sum_{j=M+1}^D \lambda_j$$

$$\mathbf{W} = \mathbf{U}_M \left(\boldsymbol{\Lambda}_M - \sigma^2 \mathbf{I} \right)^{1/2}$$



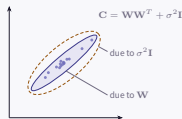
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INTUITION

Assume measured covariance $\mathbf{S}$ decomposes like

$\mathbf{S} = \mathbf{S}_{\text{true}} + \mathbf{S}_{\text{noise}}$ (using eigendecomposition)

$$\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T = \mathbf{U}_M\boldsymbol{\Lambda}_M\mathbf{U}_M^T + \mathbf{U}_-\boldsymbol{\Lambda}_-\mathbf{U}_-^T$$

Covariance of the Gaussian was $\mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2 \mathbf{I}$

$\mathbf{C}$ should match $\mathbf{S}_{\text{true}}$:

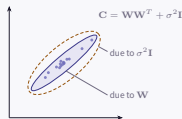
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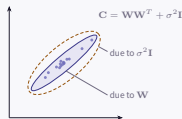
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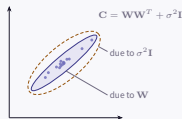
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PPCA

- PPCA is the probabilistic **generative** version of PCA: we can also draw samples from it
- PPCA is a form of Gaussian distribution with number of parameters **restricted** (by the latent space)
- PPCA is the basis of **Bayesian PCA** (Bishop 12.2.3) in which the dimension of the latent space can be found from the data (not covered)
- Like with the Gaussian Mixture Model, also the PPCA can be done via an **EM algorithm** (Bishop 12.2.2) – *Though not necessary because we have closed form solutions*

PCA: Summary

- **Three views**
 - Max variance, min reconstruction error, probabilistic
- **Applications**
 - Dimensionality reduction + representation learning / feature learning
 - 2D/3D visualization
 - Compression
 - Whitening (de-correlating features)
 - (not mentioned) De-noising: discard the smallest variance features = the noise components (hopefully!)
- **Limitation**
 - Only linear transformations